Asymptotic Behavior of the Number of Interval Sums

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Abstract

In this paper we extend the paper of O'Sullivan et al. on interval sums of prime powers to a class of regularly varying functions with index $\alpha > 1$, under mild additional assumptions. Let $n_f(x)$ be the length of the largest interval with sum less than x and $S_f(x)$ denote the number of interval sums less than x. Our main theorem states that $S_f(x) \sim n_f(x)^2 C(\alpha)$, where $C(\alpha)$ is constant dependent only on α. This paper applies tools from the theory of regularly varying functions and relates the number of blocks of length i to $\phi(x) = (1 + x^{\alpha+1})^{\frac{1}{\alpha+1}} - x$.

1 Introduction

Interval sum is the sum of consecutive terms in a sequence. Determining the asymptotic behavior of the count of interval sums less than x has been investigated for primes and prime powers [3, 4, 5]. Moser [3] determined the asymptotic behavior of interval sums of primes. Tongsomporn et al. [5] derived bounds for the number of interval sums of prime squares, though the bounds were not asymptotic to each other. O'Sullivan et al. [4] found asymptotic upper and lower bounds for the number of interval sums of prime powers.

To the best of our knowledge, no asymptotic formula is known for more general sequences. An asymptotic formula is found for the number of interval sums for a class of sequences $\big(f(i)\big)_{i=1}^{\infty}$, where f is measurable, locally bounded away from 0 and ∞, eventually increasing, and regularly varying with index $\alpha > 1$. This generalizes and extends O'Sullivan et al.'s work and provides exact asymptotics for the number of interval sums of prime powers.

Our main theorem provides an asymptotic for the interval sum counting function

$$S_f(x) = \#\left\{(i,j) \in \mathbb{N}^2 \mid i \le j, f(i) + f(i+1) + \cdots + f(j) \le x\right\}. \tag{1}$$

That is

$$S_f(x) \sim n^2 C(\alpha) \qquad n = n_f(x) \to \infty \tag{2}$$

where n is the length of the largest interval, with following asymptotic relation to x, $x \sim \frac{nf(n)}{\alpha+1}, x \to \infty$ and

$$C(\alpha) = \frac{1}{2(\alpha+1)} \frac{\Gamma(\frac{1}{\alpha+1})\Gamma(\frac{\alpha-1}{\alpha+1})}{\Gamma(\frac{\alpha}{\alpha+1})}. \tag{3}$$

The proof relies on classical results from the theory of regular variation, in particular the uniform convergence theorem and Potter's bound [1]. The function $S_f(x)$ can be split into a sum of $s_{i,f}$ that counts the number of intervals of length i. The main idea in the proof is that $s_{i,f}$ can be related to the inverse of

$$\phi(x) = (1 + x^{\alpha+1})^{\frac{1}{\alpha+1}} - x \tag{4}$$

which allows us to calculate the exact coefficient in the asymptotic.

This paper is organized as follows. Section 2 defines our notation. Section 3 first proves basic lemma about asymptotics, then a lemma that relates ϕ and $s_{i,f}$, and finally a lemma that calculates the integral of the inverse of ϕ. Section 4 proves the main theorem 1 and one technical lemma bounding the sum of $s_{i,f}$ for small i.

2 Notation

- The function $f\colon [0,\infty) \to [0,\infty)$ is assumed to satisfy following criterion: measurable, eventually increasing, locally bounded away from 0 and ∞, and regularly varying with index α (see definition 1 or [1]).
- $S_f(x) = \#\left\{(i,j) \in \mathbb{N}^2 \mid i \le j, f(i) + f(i+1) + \cdots + f(j) \le x\right\}$ – the number of interval sums below x.
- $s_{k,f}(x) = \#\left\{(i,j) \in \mathbb{N}^2 \mid j - i = k-1, f(i) + f(i+1) + \cdots + f(j) \le x\right\}$ – the number of interval sums of length k below x.
- $f(x) \sim g(x),\ x \to a$ – meaning $\lim_{x\to a} \frac{f(x)}{g(x)} = 1$. The part "$x \to a$" is often omitted when it is clear from context.
- $f(x) \lesssim g(x)$ – meaning $\limsup \frac{f(x)}{g(x)} \le 1$.
- $f(x) \gtrsim g(x)$ – meaning $g(x) \lesssim f(x)$.
- sums of the form $\sum_{i=x}^{y} F(i)$ are denoted with $\sum_{x \le i < y} F(i)$.
- $f(x) = O(g(x))$ – there exists $C > 0$ such that $\left|f(x)\right| \le Cg(x)$ for all sufficiently large x.
- $f(x) = o(g(x))$ – shorthand for

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$$

- Define $n := n_f(x)$, where

$$n_f(x) = \max \left\{ n \in \mathbb{N} \mid \sum_{i=1}^{n} f(i) \leq x \right\} \tag{5}$$

that is, length of the largest interval.

3 Preliminaries

Definition 1. Function $f\colon [0,\infty) \to [0,\infty)$ is **regularly varying** with **index** α if for all $\lambda > 0$

$$\lim_{x\to\infty} \frac{f(\lambda x)}{f(x)} = \lambda^{\alpha}. \tag{6}$$

Definition 2. Function $L\colon [0,\infty) \to [0,\infty)$ is **slowly varying** if for all $\lambda > 0$

$$\lim_{x\to\infty} \frac{L(\lambda x)}{L(x)} = 1. \tag{7}$$

Since f is measurable and regularly varying with index α, we have $f(x) = x^{\alpha}L(x)$, for slowly varying L (see Theorem 1.4.1 in [1]).

Lemma 1. *We have*

$$x \sim \frac{n^{\alpha+1}L(n)}{\alpha+1} = \frac{nf(n)}{\alpha+1} \tag{8}$$

for $\alpha > -1$. Note $n := n_f(x)$.

Proof. From Karamata's theorem (see proposition 1.5.8 in [1] or Proposition B.1 in the appendix)

$$\int_1^n f(i) \sim \frac{nf(n)}{\alpha+1}. \tag{9}$$

From eventual monotonicity and regular variation with index $\alpha > -1$ of f it follows that

$$\sum_{i=1}^{n} f(i) \sim \int_1^n f(i) \tag{10}$$

From definition of $n = n_f(x)$ the result follows. $\square$

4 Proof of the Main Theorem

Lemma 2. *Let $\varepsilon > 0$ and $1-\varepsilon \geq \lambda \geq \varepsilon$, then $s_{\lambda n,f}(x)$ has the following asymptotic uniformly*

$$s_{\lambda n,f}(x) \sim n\phi^{-1}(\lambda) \tag{11}$$

where $\phi(x) = (1+x^{\alpha+1})^{\frac{1}{\alpha+1}} - x$.

Proof. We are going to consider sums of the form $\sum_{i=ln}^{kn} f(i)$. Let $l, k > 0$ be such that $k - l = \lambda$ and $k^{\alpha+1} - l^{\alpha+1} = 1$. Such k and l exists, by monotonicity of $(l+\lambda)^{\alpha+1} - l^{\alpha+1}$. Now Using mean-value theorem we get that for some $\zeta \in (l, k)$

$$1 = (l+\lambda)^{\alpha+1} - l^{\alpha+1} = (\alpha+1)\zeta^{\alpha}\lambda > (\alpha+1)l^{\alpha}\varepsilon. \tag{12}$$

Thus

$$l < \left(\frac{1}{(\alpha+1)\varepsilon}\right)^{\frac{1}{\alpha}}. \tag{13}$$

To show that k, l are bounded below note that, if $l < \varepsilon$ then

$$1 = (l+\lambda)^{\alpha+1} - l^{\alpha+1} < (\varepsilon+\lambda)^{\alpha+1} - \varepsilon^{\alpha+1} = 1 - \varepsilon^{\alpha+1} < 1 \tag{14}$$

which is a clear contradiction.

Now k and l are in a compact subset of $(0, \infty)$, thus we can use the uniform convergence theorem (see Theorem 1.2.1 in [1] or Theorem B.1 in the Appendix). Using uniform convergence theorem and lemma 1

$$x \sim \frac{1}{\alpha+1} L(n) n^{\alpha+1} = \tag{15}$$

$$\frac{1}{\alpha+1} L(n) n^{\alpha+1}(k^{\alpha+1} - l^{\alpha+1}) = \tag{16}$$

$$L(n) \int_{nl}^{nk} t^{\alpha} dt \sim \tag{17}$$

$$L(n) \sum_{i=nl}^{nk} i^{\alpha} = \tag{18}$$

$$\sum_{i=nl}^{nk} i^{\alpha} L(n) \sim \sum_{i=nl}^{nk} f(i). \tag{19}$$

It follows that uniformly

$$x \sim \sum_{i=nl}^{nk} f(i). \tag{20}$$

Using $k^{\alpha+1} - l^{\alpha+1} = 1$ we get

$$k^{\alpha+1} - l^{\alpha+1} = 1 \tag{21}$$

$$k - l = (1 + l^{\alpha+1})^{\frac{1}{\alpha+1}} - l \tag{22}$$

$$\lambda = (1 + l^{\alpha+1})^{\frac{1}{\alpha+1}} - l \tag{23}$$

$$ln = n\phi^{-1}(\lambda). \tag{24}$$

Lemma 3 proves that ϕ^{-1} is well defined. Since $s_{\lambda n, f}(x)$ counts the number of intervals of length λn, we get from equations 20 and 24

$$s_{\lambda n, f}(x) \sim nl = n\phi^{-1}(\lambda). \tag{25}$$

Though this assumes that the inversion from $x \sim \sum_{i=nl}^{nk} f(i)$ to $s_{\lambda n,f}(x) \sim nl$ is valid uniformly. Putting everything in terms of lambda we get that uniformly

$$x \sim \sum_{i=n\phi^{-1}(\lambda)}^{n(\phi^{-1}(\lambda)+\lambda)} f(i) \tag{26}$$

Now if the convergence is not uniform there exists $\varepsilon_0 > 0$ and for all $X_0 > 0$ such that there exists $x > X_0$ and $\lambda_x \in [\varepsilon, 1-\varepsilon]$ such that

$$\left| \frac{s_{\lambda_x n,f}(x)}{n\phi^{-1}(\lambda_x)} - 1 \right| > \varepsilon_0 \tag{27}$$

Now take sequence (x_m, λ_m) that satisfies 27. There are two cases
Case 1: $s_{\lambda n,f}(x) > n\phi^{-1}(\lambda)(1+\varepsilon_0)$
It would follow that

$$x \sim \sum_{i=s_{\lambda n,f}(x)}^{s_{\lambda n,f}(x)+n\lambda} f(i) > \sum_{i=n\phi^{-1}(\lambda)(1+\varepsilon_0)}^{n(\phi^{-1}(\lambda)(1+\varepsilon_0)+\lambda)} f(i) \tag{28}$$

We also have by same argument as steps from 19 to 16 that

$$\sum_{i=n\phi^{-1}(\lambda)(1+\varepsilon_0)}^{n(\phi^{-1}(\lambda)(1+\varepsilon_0)+\lambda)} f(i) \sim x((\phi^{-1}(\lambda)(1+\varepsilon_0)+\lambda)^{\alpha+1} - ((\phi^{-1}(\lambda)(1+\varepsilon_0))^{\alpha+1}) \tag{29}$$

We know that

$$((\phi^{-1}(\lambda)(1+\varepsilon_0)+\lambda)^{\alpha+1} - ((\phi^{-1}(\lambda)(1+\varepsilon_0))^{\alpha+1}) = 1 + F(\lambda) \tag{30}$$

for some continuous F_+ and that $F_+(\lambda) > 0$, hence it has minimum $c > 0$ in the compact λ-set. Thus independent λ we get that

$$x \gtrsim x(1+c) \tag{31}$$

which is a contradiction.

Case 2: $s_{\lambda n,f}(x) < n\phi^{-1}(\lambda)(1-\varepsilon_0)$
It would follow that

$$x \sim \sum_{i=s_{\lambda n,f}(x)}^{s_{\lambda n,f}(x)+n\lambda} f(i) < \sum_{i=n\phi^{-1}(\lambda)(1-\varepsilon_0)}^{n(\phi^{-1}(\lambda)(1-\varepsilon_0)+\lambda)} f(i) \tag{32}$$

Where inequality comes from monotonicity of f. We also have by same argument as steps from 19 to 16 that

$$\sum_{i=n\phi^{-1}(\lambda)(1-\varepsilon_0)}^{n(\phi^{-1}(\lambda)(1-\varepsilon_0)+\lambda)} f(i) \sim x((\phi^{-1}(\lambda)(1-\varepsilon_0)+\lambda)^{\alpha+1} - ((\phi^{-1}(\lambda)(1-\varepsilon_0))^{\alpha+1}) \tag{33}$$

We know that

$$((\phi^{-1}(\lambda)(1-\varepsilon_0)+\lambda)^{\alpha+1}-((\phi^{-1}(\lambda)(1-\varepsilon_0))^{\alpha+1})=1-F(\lambda) \tag{34}$$

for some continuous F_- and that $F_-(\lambda)>0$, hence it has minimum $c>0$ in the compact λ-set. Thus independent λ we get that

$$x \lesssim x(1-c) \tag{35}$$

which is a contradiction. ☐

Lemma 3. *The function $\phi:(0,\infty)\to(0,1)$, $\phi(x)=(1+x^{\alpha+1})^{\frac{1}{\alpha+1}}-x$ is a strictly decreasing bijection, has an inverse, and*

$$\int_0^1 \phi^{-1}(t)dt = \frac{1}{2(\alpha+1)}\frac{\Gamma(\frac{1}{\alpha+1})\Gamma(\frac{\alpha-1}{\alpha+1})}{\Gamma(\frac{\alpha}{\alpha+1})} \tag{36}$$

for $\alpha>1$.

Proof. It is straightforward to verify that ϕ has an inverse and it's integral converges. Doing change of variables and using analytic continuation of the beta function, gives the evaluation. For full details see the appendix A. ☐

Lemma 4. *Let $\varepsilon>0$, $\alpha>1$, and $0\le l\le 1-\varepsilon$ then*

$$\sum_{i=nl}^{n(l+\varepsilon)} s_{i,f}(x) = O(n^2)\varepsilon^{\frac{1}{2}(1-\frac{1}{\alpha})}, \tag{37}$$

where the $O(n^2)$ term is independent of ε.

Proof. Since

$$x \ge f(i)+\cdots+f(i+k-1)\ge kf(i) \tag{38}$$

$$f^{-1}\left(\frac{x}{k}\right)\ge i, \tag{39}$$

it follows that

$$s_{k,f}(x)\le f^{-1}\left(\frac{x}{k}\right). \tag{40}$$

The inverse of a regularly varying function is regularly varying with index $\frac{1}{\alpha}$ (see Theorem 1.5.12 of [1] or Theorem B.3 in the Appendix). Let $f^{-1}(x)=x^{\frac{1}{\alpha}}\tilde{L}(x)$. We get from monotonicity of f that

$$\sum_{i=nl}^{n(l+\varepsilon)} f^{-1}\left(\frac{x}{i}\right)\sim\int_{nl}^{n(l+\varepsilon)} f^{-1}\left(\frac{x}{t}\right)dt = x^{\frac{1}{\alpha}}\int_{nl}^{n(l+\varepsilon)} t^{-\frac{1}{\alpha}}\tilde{L}\left(\frac{x}{t}\right)dt. \tag{41}$$

Since $t \leq n$, we get by Potter's bound (ii)

$$\tilde{L}\left(\frac{x}{t}\right) \leq M\tilde{L}\left(\frac{x}{n}\right)\left(\frac{n}{t}\right)^{\delta}, \tag{42}$$

where M is the constant from Potter's bound and $\delta = \frac{1}{2}(1 - \frac{1}{\alpha})$. Now we can bound the integral

$$x^{\frac{1}{\alpha}} \int_{nl}^{n(l+\varepsilon)} t^{-\frac{1}{\alpha}} \tilde{L}\left(\frac{x}{t}\right) dt \leq x^{\frac{1}{\alpha}} M \tilde{L}\left(\frac{x}{n}\right) n^{\delta} \int_{nl}^{n(l+\varepsilon)} t^{-\frac{1}{\alpha}-\delta}. \tag{43}$$

By definition

$$f^{-1}(f(n)) \sim n \tag{44}$$
$$n\tilde{L}(n^{\alpha}L(n))L(n)^{1/\alpha} \sim n \tag{45}$$
$$\tilde{L}(n^{\alpha}L(n))L(n)^{1/\alpha} \sim 1. \tag{46}$$

Using the relation between x and n obtained in Lemma 1

$$\tilde{L}\left(\frac{x}{n}\right) \sim \tilde{L}\left(n^{\alpha}L(n)\right). \tag{47}$$

Thus, from equations 47 and 46

$$x^{\frac{1}{\alpha}} M \tilde{L}\left(\frac{x}{n}\right) n^{\delta} \int_{nl}^{n(l+\varepsilon)} t^{-\frac{1}{\alpha}-\delta} \tag{48}$$
$$\sim Mn^2 L(n)^{\frac{1}{\alpha}} \tilde{L}(n^{\alpha}L(n)) \frac{2\alpha}{\alpha-1}((l+\varepsilon)^{\frac{1}{2}(1-\frac{1}{\alpha})} - l^{\frac{1}{2}(1-\frac{1}{\alpha})}) \tag{49}$$
$$\sim Mn^2 \frac{2\alpha}{\alpha-1}((l+\varepsilon)^{\frac{1}{2}(1-\frac{1}{\alpha})} - l^{\frac{1}{2}(1-\frac{1}{\alpha})}) \tag{50}$$
$$= O(n^2)((l+\varepsilon)^{\frac{1}{2}(1-\frac{1}{\alpha})} - l^{\frac{1}{2}(1-\frac{1}{\alpha})}) \tag{51}$$
$$= O(n^2)\varepsilon^{\frac{1}{2}(1-\frac{1}{\alpha})}. \tag{52}$$

□

Theorem 1. *Under the assumptions and notation in 2, the function $S_f(x)$ has the asymptotic $n_f(x)^2 C(\alpha), x \to \infty$ for $\alpha > 1$, where*

$$C(\alpha) = \frac{1}{2(\alpha+1)} \frac{\Gamma(\frac{1}{\alpha+1})\Gamma(\frac{\alpha-1}{\alpha+1})}{\Gamma(\frac{\alpha}{\alpha+1})}. \tag{53}$$

Proof. Let $\varepsilon > 0$. From the definition of S and $s_{i,f}$ it follows that

$$S_f(x) = \sum_{i=1}^{n} s_{i,f}(x). \tag{54}$$

Splitting the sum

$$S_f(x) = \sum_{i=1}^{n\varepsilon} s_{i,f}(x) + \sum_{i=n\varepsilon}^{n(1-\varepsilon)} s_{i,f}(x) + \sum_{i=n(1-\varepsilon)}^{n} s_{i,f}(x). \tag{55}$$

Using Lemma 2 and continuity of ϕ^{-1}, we get

$$\begin{aligned}
\sum_{i=n\varepsilon}^{n(1-\varepsilon)} s_{i,f}(x) &\sim \sum_{i=n\varepsilon}^{n(1-\varepsilon)} n\phi^{-1}\left(\frac{i}{n}\right) && (56)\\
&= n \sum_{i=n\varepsilon}^{n(1-\varepsilon)} \phi^{-1}\left(\frac{i}{n}\right) && (57)\\
&\sim n \int_{n\varepsilon}^{n(1-\varepsilon)} \phi^{-1}\left(\frac{t}{n}\right) dt && (58)\\
&= n^2 \int_{\varepsilon}^{1-\varepsilon} \phi^{-1}(t)dt. && (59)
\end{aligned}$$

Since the integral converges

$$\lim_{\varepsilon\to 0^+} \int_{\varepsilon}^{1-\varepsilon} \phi^{-1}(t)dt = \int_0^1 \phi^{-1}(t)dt \tag{60}$$

From Lemma 4, it follows that the first and third sum in 55 are negligible in the asymptotic, hence taking $\varepsilon \to 0$ finishes the proof. $\square$

Corollary 1. *The number of interval sums for $f(x) = x^\alpha, \alpha > 1$ is asymptotic to*

$$x^{\frac{2}{1+\alpha}}(\alpha+1)^{\frac{2}{1+\alpha}}C(\alpha) \tag{61}$$

Proof. From theorem 1 and inverting lemma 1, it follows that

$$S_f(x) \sim n^2C(\alpha) \sim ((\alpha+1)^{\frac{1}{1+\alpha}}x^{\frac{1}{1+\alpha}})^2C(\alpha) \sim x^{\frac{2}{1+\alpha}}(\alpha+1)^{\frac{2}{1+\alpha}}C(\alpha) \tag{62}$$

$\square$

Corollary 2. *Number of interval sums for prime powers $p^\alpha, \alpha > 1$ is asymptotic to*

$$(1+\alpha)^2C(\alpha)\left(\frac{x}{\log(x)^\alpha}\right)^{\frac{2}{\alpha+1}} \tag{63}$$

Proof. From the prime number theorem, it follows that $f(x) \sim (x\log(x))^\alpha$. Now using lemma 1 it follows that $x \sim \frac{1}{\alpha+1}n^{\alpha+1}\log^\alpha(n)$. It is straightforward to verify that $n = (1+\alpha)\left(\frac{x}{\log^\alpha(x)}\right)^{\frac{1}{1+\alpha}}$ is the asymptotic inverse. By theorem 1 the corollary follows. $\square$

5 Convergence of the Asymptotic Formula

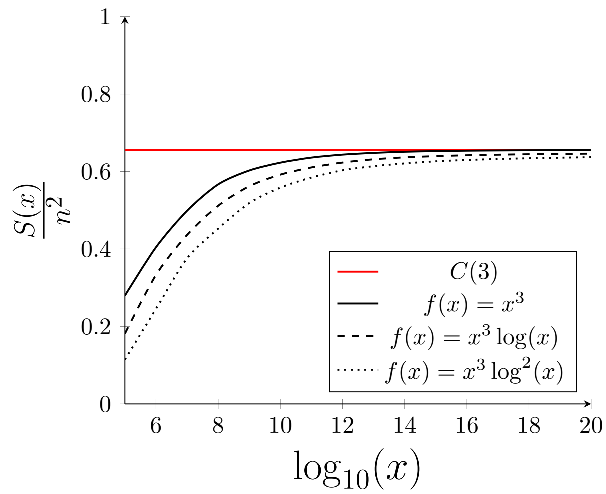

Figure 1: Example of how the slowly varying part affects convergence

Figure 1 shows the impact of the slowly varying part. We can observe that larger growth rate in the slowly varying part corresponds to slower convergence.

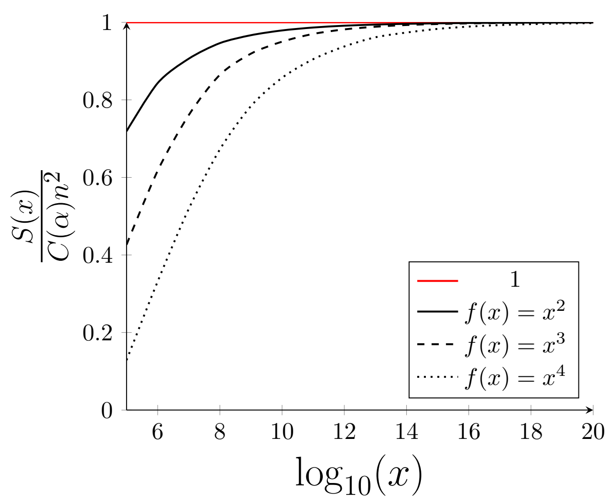

Figure 2: Example of how the index affects convergence

Figure 2 shows the impact of the index on convergence. We can observe that increasing the index slows down the convergence.

See the table 1 below for relative error of the asymptotic at $x = 10^{20}$ for varying functions.

$f(x)$	relative error at $x = 10^{20}$
x^2	0.037%
x^3	0.045%
x^4	0.194%
x^5	0.771%
$x^3 \log(x)$	1.433%
$x^3 \log(x)^2$	2.836%
$x^4/\sqrt{\log(x)}$	0.215%
$x^4/\log(x)$	0.622%

Table 1: Relative error of the asymptotic at $x = 10^{20}$ for various functions.

These findings align with our analysis: the proofs rely on the uniform convergence theorem, which gives slower convergence for larger growth rates, and the convergence of the sum to integral is slower for larger indices.

6 Conclusion

The main theorem gives an asymptotic for the number of interval sums below x, namely $S_f(x) \sim n^2 C(\alpha)$. It remains an open question what happens in the case $\alpha \leq 1$; our methods can't handle that case since the integral in Lemma 3 diverges. Future work could therefore investigate the case $\alpha \leq 1$, it seems fruitful to use methods of de Haan theory to investigate the case $\alpha = 1$.

A Appendix: The Integral of the Function Phi

Here we give detailed evaluation of the integral of ϕ.

Proof. Taking the derivative of ϕ

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = (1 + x^{\alpha+1})^{-\frac{\alpha}{\alpha+1}} x^{\alpha} - 1 < 0 \tag{64}$$

As $\phi(0) = 1$ and $\lim_{x\to\infty} \phi(x) = 0$, ϕ is a bijection, hence it has an inverse. Since ϕ is a strictly decreasing bijection

$$\int_0^1 \phi^{-1}(x)dx = \int_0^\infty \phi(x)dx. \tag{65}$$

The integral converges as

$$\phi(x) = (1 + x^{\alpha+1})^{\frac{1}{\alpha+1}} - x < \frac{1}{1+\alpha} x^{-\alpha}. \tag{66}$$

Doing a change of variables $t = \frac{x^{\alpha+1}}{1+x^{\alpha+1}}$, $x = (\frac{t}{1-t})^{\frac{1}{\alpha+1}}$, we get

$$\int_0^\infty \phi(x)dx = \int_0^1 (1-t)^{-\frac{1}{\alpha+1}}(1-t^{\frac{1}{\alpha+1}})\frac{1}{\alpha+1}\left(\frac{t}{1-t}\right)^{-\frac{\alpha}{\alpha+1}}(1-t)^{-2}dt \quad (67)$$

$$= \frac{1}{\alpha+1}\int_0^1 (1-t)^{-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})dt. \quad (68)$$

Let $g_r(t) = (1-t)^{r-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})$ and $g(t) = (1-t)^{-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})$. Since for each $r > 0$, $g_r(t) \to g(t), r \to 0$ pointwise and $|g_r(t)| < |g(t)|$, we can use the dominated convergence theorem. Thus

$$\int_0^1 (1-t)^{-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})dt = \lim_{r\to 0^+}\int_0^1 (1-t)^{r-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})dt. \quad (69)$$

Let B be the beta function (see NIST DLMF § 5.12.1 [2])

$$B(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1}dt = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1+z_2)}. \quad (70)$$

The beta function can be analytically extended using Pochhammer's integral (see NIST DLMF § 5.12.12 [2]).

$$\int_0^\infty \phi(x)dx = \frac{1}{\alpha+1}\int_0^1 (1-t)^{-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})dt \quad (71)$$

$$= \frac{1}{\alpha+1}\lim_{r\to 0^+}\int_0^1 (1-t)^{r-\frac{2}{\alpha+1}-1}(t^{-\frac{\alpha}{\alpha+1}} - t^{\frac{1-\alpha}{\alpha+1}})dt \quad (72)$$

$$= \frac{1}{\alpha+1}\lim_{r\to 0^+}\left(B\left(\frac{1}{\alpha+1}, r - \frac{2}{\alpha+1}\right) - B\left(\frac{2}{\alpha+1}, r - \frac{2}{\alpha+1}\right)\right). \quad (73)$$

Using the gamma expression for beta functions, we get

$$\int_0^\infty \phi(x)dx = \frac{1}{\alpha+1}\lim_{r\to 0^+}\left(\frac{\Gamma(\frac{1}{\alpha+1})\Gamma(r-\frac{2}{\alpha+1})}{\Gamma(\frac{1}{\alpha+1}+r-\frac{2}{\alpha+1})} - \frac{\Gamma(\frac{2}{\alpha+1})\Gamma(r-\frac{2}{\alpha+1})}{\Gamma(\frac{2}{\alpha+1}+r-\frac{2}{\alpha+1})}\right) \quad (74)$$

$$= \frac{1}{\alpha+1}\lim_{r\to 0^+}\left(\frac{\Gamma(\frac{1}{\alpha+1})\Gamma(-\frac{2}{\alpha+1})}{\Gamma(-\frac{1}{\alpha+1})} - \frac{\Gamma(\frac{2}{\alpha+1})\Gamma(-\frac{2}{\alpha+1})}{\Gamma(r)}\right). \quad (75)$$

Since $\Gamma(t) = \frac{\Gamma(1+t)}{t}$, we have $\Gamma(t) \sim \frac{1}{t},\ t \to 0$. Thus

$$\lim_{r\to 0}\frac{\Gamma(\frac{2}{\alpha+1})\Gamma(-\frac{2}{\alpha+1})}{\Gamma(r)} = 0 \quad (76)$$

so the expression simplifies to

$$\int_0^\infty \phi(x)dx = \frac{1}{\alpha+1}\frac{\Gamma(\frac{1}{\alpha+1})\Gamma(-\frac{2}{\alpha+1})}{\Gamma(-\frac{1}{\alpha+1})} \tag{77}$$

$$= \frac{1}{2(\alpha+1)}\frac{\Gamma(\frac{1}{\alpha+1})\Gamma(\frac{\alpha-1}{\alpha+1})}{\Gamma(\frac{\alpha}{\alpha+1})} \tag{78}$$

where the last equality follows from $z\Gamma(z) = \Gamma(z+1)$. □

B Appendix: Selected Results on Regular Variation

For convenience, we reproduce several standard theorems from [1] we assume through this section that f and l are measurable.

Theorem B.1 (Uniform Convergence Theorem)**.** *If l is slowly varying then*

$$\frac{l(\lambda x)}{l(x)} \to 1 \qquad (x \to \infty)$$

uniformly in each compact λ-set in $(0, \infty)$.

Proof. See Theorem 1.2.1 in [1]. □

Theorem B.2 (Potters Theorem)**.** *(i) If l is slowly varying function then for any chosen constants $A > 1$, $\delta > 0$ there exists $X = X(A, \delta)$ such that*

$$\frac{l(y)}{l(x)} \leq A \max\left\{\left(\frac{y}{x}\right)^\delta, \left(\frac{x}{y}\right)^\delta\right\} \qquad (x \geq X, y \geq X)$$

(ii) If further, l is bounded away from 0 and ∞ in every compact subset of $[0, \infty)$, then for every $\delta > 0$ there exists $A' = A'(\delta) > 1$ such that

$$\frac{l(y)}{l(x)} \leq A' \max\left\{\left(\frac{y}{x}\right)^\delta, \left(\frac{x}{y}\right)^\delta\right\} \qquad (x \geq 0, y \geq 0)$$

(iii) If f is regularly varying of index ρ then for any chosen $A > 1$, $\delta > 0$ there exists $X = X(A, \delta)$ such that

$$\frac{f(y)}{f(x)} \leq A \max\left\{\left(\frac{y}{x}\right)^{\rho+\delta}, \left(\frac{x}{y}\right)^{\rho+\delta}\right\} \qquad (x \geq X, y \geq X)$$

Proof. See Theorem 1.5.6 in [1]. □

Proposition B.1 (Karamata's Theorem)**.** *If l is slowly varying, X is so large that $l(x)$ is locally bounded in $[X, \infty)$, and $\alpha > -1$, then*

$$\int_X^x t^\alpha l(t)dt \sim \frac{x^{\alpha+1}l(x)}{\alpha+1} \qquad (x \to \infty)$$

Proof. See Proposition 1.5.8 in [1]. □

Definition B.1 (Generalized inverse)**.** Generalized inverse of f is defined by

$$f^{\leftarrow}(x) = \inf\left\{x \mid f(x) = y\right\} \tag{79}$$

Theorem B.3. *If $f \in RV_\alpha$ with $\alpha > 0$, there exits $g \in RV_{\frac{1}{\alpha}}$ with*

$$f(g(x)) \sim g(f(x)) \sim x \qquad (x \to \infty).$$

Here g (an asymptotic inverse of f) is determined uniquely to within asymptotic equivalence, and one version of g is $f^{\leftarrow}$.

Proof. See theorem 1.5.12 in [1]. □

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